

Object Oriented Programming (OOP)

“Lab 06”

(Constructors and Destructors in Inheritance)

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Course Code: CSC-521

Task 1: Online Shopping System (Constructor):

Create a base class **Product** with properties **name** and **price**. Create a derived class **Electronics** with property **warrantyYears**. Use constructor chaining to initialize all values and display them.

Task 2: Ride Booking System (Constructor - Multi-level):

Create classes **Vehicle** -> **Car** -> **ElectricCar**. Each class should have a constructor printing a message. Pass values through constructors and observe execution order.

Task 3: File Handling System (Destructor):

Create a base class **File** that prints '**File opened**' in constructor and '**File closed**' in destructor.

Create a derived class **TextFile** that prints its own messages. Create object and observe destructor order.

Task 4: Device and Sensor System (Destructor with Dynamic Memory):

Create base class **Device** with properties **deviceName**, **deviceId**, and dynamically allocated **data**.

Create derived class **Sensor** with **sensorType** and dynamically allocated **sensorValue**. Use constructors to allocate memory and destructors to free memory. Test using **new** and **delete**.
